



Sandipan Chattopadhyay, CTO, Justdial

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technology anywhere in the world. It was going to be the first of its kind. It always felt like the tech demands within the firm were cropping up with great urgency, continuously, and these demands were growing. And hence, going into something that was a closed box would dampen the firm’s innovativeness in some way or the other. So the only solution was open source, and it was not just about taking bits and pieces of it but doing open source all the way.”

Not only at the infrastructure level, but also at the user level, Sandipan and team have really been able to take open source technology to an entirely new level at Justdial. He explains, “There are quite a few things we have done in Justdial, with respect to open source technology that we are really proud of. One of the things is that a majority of our workforce doesn’t see a Windows box. Even the desktops used at Justdial, which number about

5000-6000, all run on Ubuntu’s home grown flavour. The Ubuntu that we use is optimised from a technical perspective, depending on the work and purpose of a particular desktop.”

Customisation is the key

Justdial’s open source leader has harnessed the customisability of the software to a great extent. He has customised technology for each function. Sandipan explains, “We have different functions being performed at Justdial that include marketing, answering customer calls, etc. Each function has different requirements from the machine. For the person who is on a calling function, what matters is that data caching is better, the response time of a Java script is faster, et al. While for the telecom people who work on diallers, what matters are the Ubuntu customisations that have resulted in the auto-dialling component, the component that takes or allocates the work, the one that does the local database lookout to find matches to suggest to the client, etc. So the way we bundle the kernel to give priority and the priority queue to different child objects is what we change.

“More importantly, in order to enhance the performance, we had to strip out things from the basic kernel that we did not need. So, if you don’t want to allow the user to watch a video, don’t preload the components to handle high multimedia content. That is how the kernel is bundled and made compatible to the particular usage cases. So we have customised the software for particular functions, objectives and use cases.”

Technology revamped in a unique way

At the time when Sandipan joined Justdial, there was not much happening with respect to technology within the company, leave alone open source technology. He changed the scenario over a period of time. He says, “There was not much technology being used in many areas at Justdial. Currently, marketing guys carry their laptops and they are more of the advocates of the software from the clients’ perspective. It was more about manual processes at that time. So some of the things that we are talking of right now didn’t even exist at that point in time. Frankly, the way we have been able to deploy and develop solutions so fast on open source has improved the adoption of technology at Justdial.”

Everything in Justdial is happening on software that is home grown and is based on the open source stack. They call it the ‘CLAMP’ stack, which is a slight deviation from the LAMP stack in the sense that the firm uses a lot of C modules as well. Sandipan explains the technology, “What is important to focus on is the architecture. It is a distributed environment because we have eight offices across the country and we cater to a huge volume of search requests. The search complexity that is a result of our business requirements and the sensitivity of the local searches is of a slightly higher level. Despite that, the kind of performance or the turnaround time that we give is the main factor for us. Everything has to be really fast.

“At the same time, the system is not typical program